Jahangirnagar University
জাহাঙ্গীরনগর বিশ্ববিদ্যালয়

Department of Computer Science and Engineering

Submission Date: 9 / 9 / 25

2nd Year 1st Semester (Hons)

Course Title: Digital Logic Design Laboratory

Course Code: CSE-254

Experiment No: 07

Experiment Title: Implementation of a boolean expression using a 4:1 multiplexer.

Submitted To:

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Batch:	52

Experiment No: 07

Experiment Name: Implementation of a boolean expression using a 4:1 multiplexer.

Objective:

- To implement the given boolean expression
- To implement a 4:1 multiplexer.

Required components:

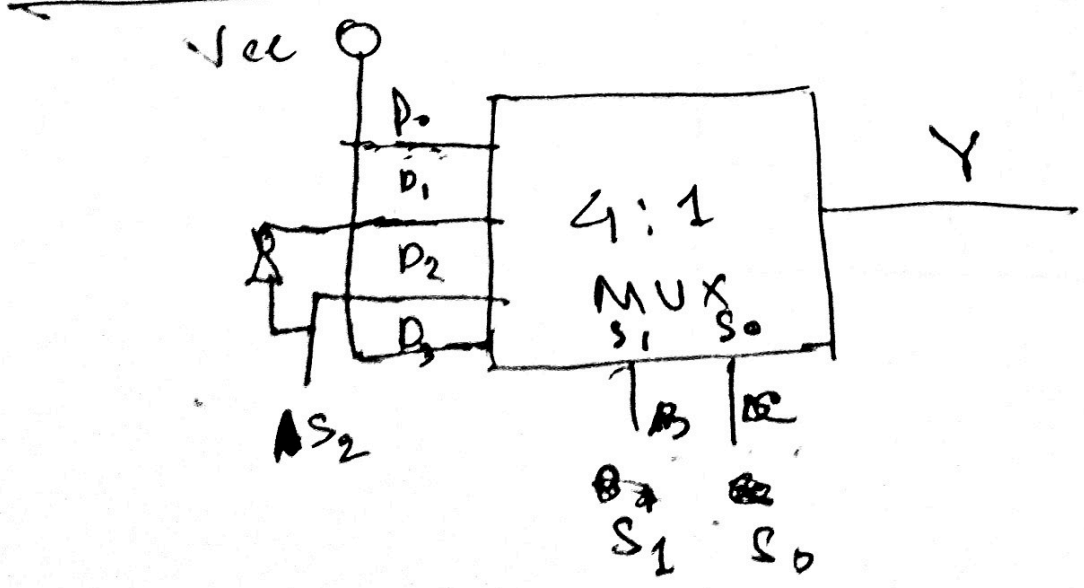
Digital trainer board, breadboard, IC 7404, IC 7408, IC 7432, connecting wires.

Theory:

Given boolean expression, $Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + AB\bar{C} + ABC$

A 4:1 MUX has 4 data inputs D_0, D_1, D_2, D_3, 2 select lines and 1 output (Y).

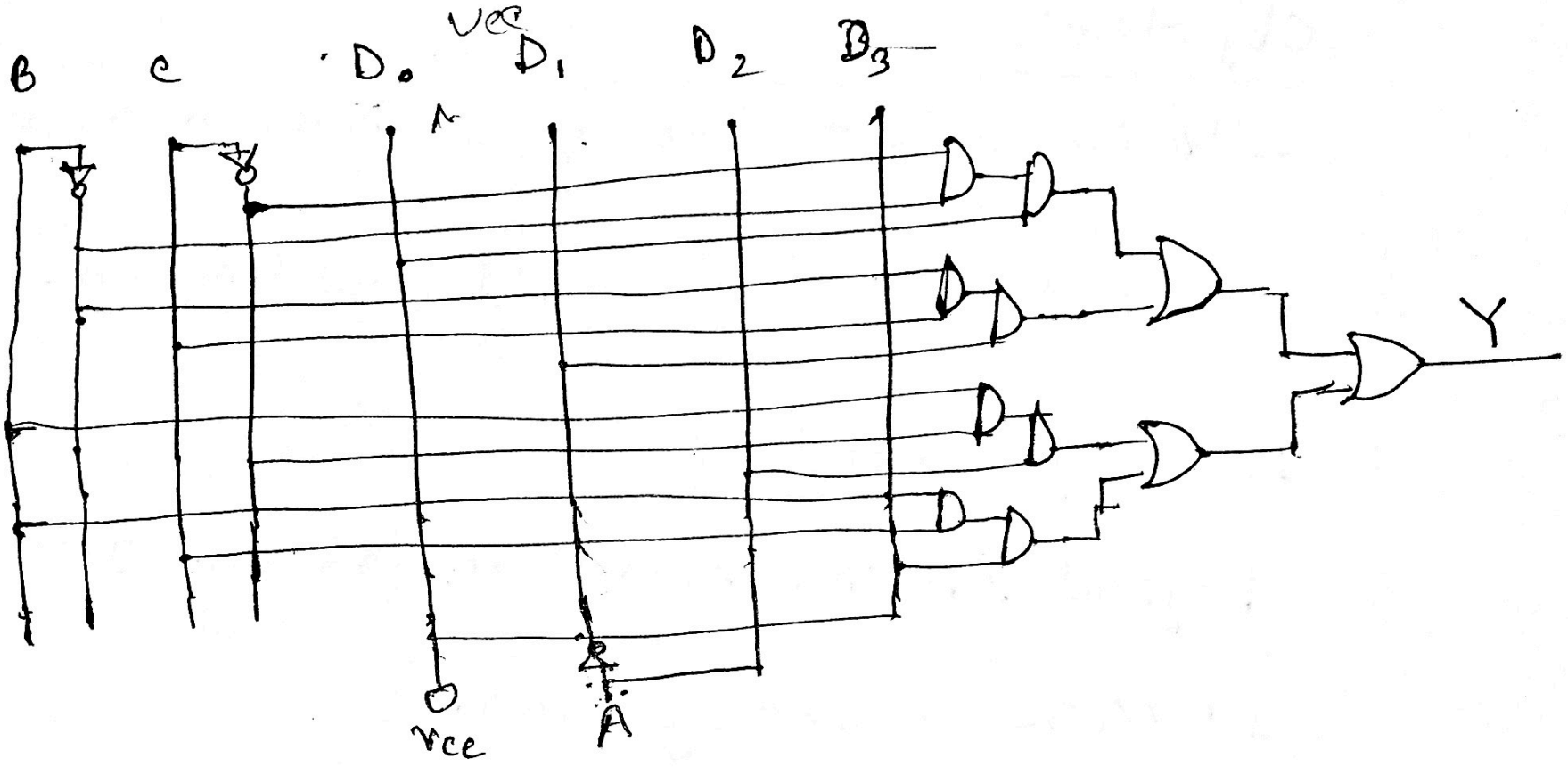
Block Diagram:



Circuit Diagram:

$$Y = \cancel{\bar{B}\bar{C}}\, D_0 + \bar{B}C D_1 + B\bar{C} D_2 + BC D_3$$

$$= \bar{B}\bar{C} \cdot V_{cc} + \bar{B}C\bar{A} + B\bar{C}A + BC \cdot V_{cc}$$

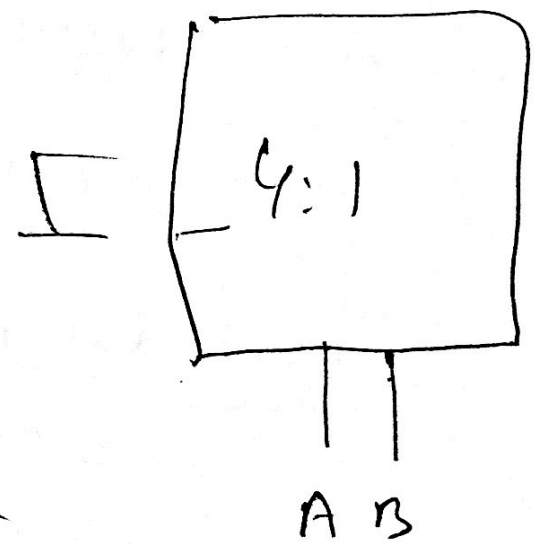


Truth Table:

A	B	C	Y	
0	0	0	1	$D_0 - Vcc$
0	0	1	1	
0	1	0	0	D_1 C
0	1	1	1	
1	0	0	1	$D_2 \cdot \bar{C}$
1	0	1	0	
1	1	0	1	$D_3 - Vcc$
1	1	1	1	

Ravi 02-09-25

A / $\bar{A}$



Result:

- The output Y matched the truth table for all input combinations.
- The LED lit up for Y=1 and remained off for Y=0.

Analysis:

The 4:1 MUX successfully implemented the expression. Practical results perfectly matched theoretical expectations.